Yi Xia (Sean)

AI Engineer · Computer Vision Researcher · PhD Candidate

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SUMMARY

Computer-vision researcher and AI engineer based in Auckland. Full-time AI Developer at TOHU MEDIA, a Māori-owned media-production company, where I design and ship object-detection and OCR systems for film-industry video footage. Concurrently completing my PhD at Auckland University of Technology on graph neural networks for skeleton-based human activity recognition; first-author paper accepted at Pattern Recognition (Q1, 2026). Six peer-reviewed papers, 70+ citations on Google Scholar, four years of industry experience earned alongside postgraduate study, and eight years of cumulative classroom teaching. Looking for a senior Computer Vision / Machine Learning Engineer role with a New Zealand team building practical AI products.

SKILLS

Programming: Python (expert), Java.

Deep learning: PyTorch (primary), TensorFlow, Keras, OpenCV, scikit-learn, NumPy, pandas, Hugging Face.

Computer vision: object detection (incl. YOLO family), object tracking, action recognition, OCR, semantic and instance segmentation, classification, video analytics.

Research methods: graph convolutional networks, attention and Transformers, skeleton-based human activity recognition, frequency-domain learning.

Generative AI and LLM: LLM integration, agentic workflows, agent skills and tool use, Claude / GPT APIs, prompt engineering.

Deployment: Docker, containerised model inference on cloud-hosted GPU servers.

Tooling: Git, GitHub, Linux, CUDA, Jupyter.

EXPERIENCE

Artificial Intelligence Developer — **Feb 2022 – present**

TOHU MEDIA, Auckland, New Zealand

Full-time since June 2025; previously part-time alongside MSc and PhD.

- AI engineer at a Māori-owned media-production company, responsible for the company's AI capability end-to-end. Design and ship object-detection and OCR systems that operate on client video footage inside internal production pipelines (specific deliverables under NDA).
- Engineer for the messy edge cases of real footage that off-the-shelf computer-vision models stumble on: variable cinematography and framing, dense overlay text in non-standard layouts, archival material with degraded contrast, and long-form takes where naive per-frame inference doesn't fit a single-workstation GPU budget. Pick model architecture, training regime, and post-processing accordingly.
- Own the full machine-learning lifecycle: dataset scope and labelling spec, model training and evaluation in PyTorch, Docker-packaged inference deployed to the company's cloud GPU servers, and iteration with editors and the creative director on what "useful output" looks like inside an actual editing workflow.

- Make independent calls on architecture, latency / accuracy trade-offs, and scope reduction under deadline, across four years part-time alongside the MSc and PhD, and now full-time ownership of the AI work.

PhD Researcher **Oct 2023 – Oct 2026**

Institute of Robotics and Vision (IoRV), Auckland University of Technology

Supervisors: Dr Raymond Lutui (primary), Associate Professor Sira Yongchareon.

- Designed FATL-GCN, a frequency-aware spatio-temporal graph network for skeleton-based human activity recognition; outperforms prior state-of-the-art on NTU RGB+D 60 / 120. Accepted at Pattern Recognition (Q1 journal, IF = 7.6) in 2026.
- Authored Dynamic Self-Attention Gated ST-GCN at ICONIP 2024; introduces learned gating between spatial and temporal streams.
- Full ownership of first-author research, from idea to model design, GPU training, ablations, paper writing and rebuttal.
- Mentor master's students on PyTorch, experimental hygiene, and academic writing.

Research Assistant **Nov 2022 – Jul 2023**

School of Engineering, Computer and Mathematical Sciences, Auckland University of Technology

- Funded research assistant on the Kiwifruit detection and counting research line. Co-built KiwiDetector (improved YOLOv7) and KiwiTracker for real-time orchard video; the lead paper has 50+ citations.
- Co-authored three peer-reviewed papers (IVCNZ 2022, SAI Intelligent Systems 2023, PSIVT 2023) during the assistantship.

Teaching Assistant **2022 – present**

Auckland University of Technology

- Teaching assistant on multiple undergraduate computer-science courses (programming, computer vision, artificial intelligence). Run labs, mark assignments, hold consultation hours, and mentor on final-year projects.

Tech Service Specialist **Apr 2023 – Dec 2023**

PB Tech, Auckland, New Zealand

- Diagnosed hardware and software issues and supported walk-in customers in a busy retail tech-service environment; sharpened plain-English technical communication with non-specialist users.

Co-Founder **2019 – 2021**

Baiyin Pinuo Training School, China

- Co-founded a private training school. Built curriculum, ran operations and finance, then divested in 2021 to relocate to New Zealand for postgraduate study; the school remains in operation under my co-founder.

Junior Lecturer **2015 – 2019**

Department of Information Engineering, Baiyin Vocational and Technology College, China

- Taught Computer Fundamentals, Java, IoT applications and RFID technologies. Four years of full-classroom teaching to approximately 120 students per year.

EDUCATION

PhD, Computer and Information Sciences **Oct 2023 – Oct 2026 (expected)**

Auckland University of Technology, New Zealand

Thesis on deep learning for skeleton-based human activity recognition. Funded by the AUT PhD Full Scholarship (tuition and stipend).

Master of Computer and Information Sciences **2021 – 2023**

Auckland University of Technology, New Zealand

Specialisation: Computer Vision.

Bachelor of Engineering, Logistics Engineering (Software Development) **2011 – 2015**

Dalian Neusoft University of Information, China

AWARDS AND HONOURS

- AUT PhD Full Scholarship — tuition plus stipend, October 2023 – October 2026.
- SECMS Research Assistant Fund, 2023.
- AUT Summer Research Assistant Scholarship, 2022.

ACADEMIC SERVICE

- Peer reviewer, ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), 2024.
- Peer reviewer, International Agrophysics, 2025 and 2026.
- Peer reviewer, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), 2026.

SELECTED PUBLICATIONS

Six peer-reviewed papers, 70+ citations; full list on Google Scholar.

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition." Pattern Recognition (Q1 journal). https://doi.org/10.1016/j.patcog.2026.113146

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). "Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR." ICONIP 2024, Springer LNCS. https://doi.org/10.1007/978-981-96-6588-4_14

Y. Xia, M. Nguyen, W. Q. Yan (2025). "An Improved YOLO Model for Kiwifruit Detection." Optimization, Machine Learning, and Fuzzy Logic, IGI Global book chapter. https://doi.org/10.4018/979-8-3693-7352-1.ch008

Y. Xia, M. Nguyen, W. Q. Yan (2022). "A Real-Time Kiwifruit Detection Based on Improved YOLOv7." IVCNZ 2022, Springer LNCS. (50 citations) https://doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, M. Nguyen, W. Q. Yan (2023). "Kiwifruit Counting Using KiwiDetector and KiwiTracker." SAI Intelligent Systems Conference, Springer. https://doi.org/10.1007/978-3-031-47724-9_41

Y. Xia, M. Nguyen, R. Lutui, W. Q. Yan (2023). "Multiscale Kiwifruit Detection from Digital Images." PSIVT 2023. https://doi.org/10.1007/978-981-97-0376-0_7

REFERENCES

Available on request. Doctoral supervisors (Dr Raymond Lutui, AUT; Associate Professor Sira Yongchareon, AUT) and an industry referee from TOHU MEDIA can be provided.